

Win-Reason & Winner Evaluation vs Ground Truth

Stage 6 (part 2) — Evaluation Against Ground Truth

Files: `win_predictor/evaluate_predictions.py` (evaluation script), `vk15b2/vk15b_dataset/<match>/<match>.csv` (ground truth).

This documents how `win_predictions.csv` output is scored against real ground truth, and the current measured accuracy across all 8 labeled matches in `vk15b_dataset/`.

Ground truth source

`vk15b2/vk15b_dataset/` contains 8 matches (`Test1_Full`, `Test2_Full`, `Test3_Full`, `Test4_Full`, `Test6_Full`, `Test7_Full`, `Test9_Full`, `Test10_Full`), each with a hand-labeled `<match>.csv`:

```
start_time,end_time,win_point_player,win_reason,ball_types,lose_reason,roundscore_A,roundscore_B
0:08,0:24,Player B,Opponent Out of Bounds,Lift Shot,Hit Out of Bounds,0,1
```

`win_reason` uses free-text labels, mapped to the pipeline's 3-class scheme with the same mapping `win_predictor/evaluate_rule.py` uses:

GT label	Pipeline class
Opponent Out of Bounds	<code>out_of_bounds</code>
Opponent Hit the Net	<code>hits_net</code>
Winner Shot	<code>wins_by_landing</code>
Opponent Failed to Return	<code>wins_by_landing</code>

Each match folder also has the supporting files `predict.py` needs (`<match>_ball_filled.csv`, `<match>_court.json`, `<match>_rally.csv`, `player_detections.csv`), so `win_predictions.csv` can be regenerated fresh with the current code at any time — see [06_win_prediction.md](#) for the run command.

Matching predicted rallies to GT rallies

Predicted and GT rally counts don't line up exactly — rally detection sometimes finds a different number of boundaries than the human-labeled GT (e.g. Test9: 88 GT rallies vs 82 detected). Row-index comparison would silently misalign every rally after the first mismatch, so rallies are matched by **time overlap** instead:

```
IoU = overlap(pred_time, gt_time) / union(pred_time, gt_time)
```

Each predicted rally is greedily paired with the best-overlapping *unused* GT rally, only if `IoU ≥ 0.3` (same threshold used by the rally-detection evaluator, `analysis/Evaluate_rallies.py`). A predicted or GT rally with no match above that threshold is dropped from the accuracy calculation entirely — it's a *rally-detection* miss, not a winner/reason classification error, so it shouldn't be scored as one.

For each matched pair, two fields are compared directly: - `winner_correct = pred['winner'] == gt['win_point_player']` - `reason_correct = pred['win_reason'] == mapped_gt_reason`

Coverage (8 matches, current run)

Match	GT rallies	Predicted	Matched (IoU≥0.3)	Unmatched GT	Unmatched pred
Test1_Full	36	36	36	0	0
Test2_Full	75	73	72	3	1
Test3_Full	22	19	19	3	0
Test4_Full	45	41	39	6	2
Test6_Full	42	42	39	3	3
Test7_Full	31	30	30	1	0
Test9_Full	88	82	74	14	8
Test10_Full	101	93	90	11	3

399 total matched rallies used for the accuracy numbers below.

Results (current pipeline code, all 8 matches on clean video)

Metric	Accuracy
Win-reason accuracy	85.7% (342/399)

Metric	Accuracy
Winner accuracy	68.9% (275/399)

Win-reason confusion matrix (rows = GT, cols = predicted)

GT Pred	OOB	HN	WBL	Class acc
OOB	141	3	2	96.6%
HN	7	102	11	85.0%
WBL	12	7	99	83.9%

Accuracy by decision path (reason_src)

Path	What it means	Rallies	Winner acc	Reason acc
ML_RF	Trajectory RF called it directly	283	70.7%	84.8%
ML_HN	CNN net-hit gate fired (standard threshold)	23	69.6%	78.3%
ML_HN_I	CNN net-hit gate fired via the lowered net-zone threshold (Strategy I)	90	63.3%	93.3%
ML_STRAT_F	RF said WBL but rule engine + fast lateral velocity overrode to OOB	3	66.7%	0.0%

Per-match

Match	N	Winner acc	Reason acc
Test1_Full	36	77.8%	100.0%
Test2_Full	72	65.3%	93.1%
Test3_Full	19	89.5%	78.9%
Test4_Full	39	61.5%	84.6%
Test6_Full	39	82.1%	92.3%
Test7_Full	30	66.7%	76.7%
Test9_Full	74	58.1%	77.0%

Match	N	Winner acc	Reason acc
Test10_Full	90	71.1%	83.3%

Failure attribution (winner errors only)

Of the 124 rallies where the predicted winner was wrong: - 31% (39) had the wrong `win_reason`, which propagated to the wrong winner - 69% (85) had the *correct* `win_reason` but the fault-side/winner call was still wrong — a separate, not-yet-fixed problem (see below)

Strategy E removal (fixed — see `predict.py`)

The pipeline previously had a “Strategy E” override: when the CNN’s net-hit gate fired (`cnn_hn >= eff_hn_thr`), it could still be overridden back to `wins_by_landing` if the trajectory RF said `wins_by_landing` and the shuttle’s last position had positive court margin.

This was measured live and found to be actively harmful: it fired on ~90/380 rallies and was correct only ~8% of the time, collapsing HN-class accuracy to 20%. Root cause: the trajectory RF is trained as a binary `out_of_bounds` vs `wins_by_landing` classifier only (HN excluded from training — see [06_win_prediction.md](#)), so for a genuine net-hit rally it is structurally incapable of saying “hits_net” and is forced to guess between the two classes it knows. Net-hit shuttles typically land just inside the court near the net, so `margin_cm > 0` was true for most real HN rallies too — the override fired on real net-hits almost as often as on genuine misclassifications.

The 81.8%/~69% figures previously cited for this strategy (e.g. in `win_predictor/README.md`, `INSTALL_AND_RUN.md`) came from `vk15b2/netcnn/traj_classifier.py`’s offline LOMO-CV benchmark, which builds its dataset by filtering to `GT ∈ {out_of_bounds, wins_by_landing}` **before** training or predicting — so the trajectory prediction is `None` for every real HN rally in that benchmark, by construction, using information (the true label) that isn’t available at real inference time. That’s why the benchmark showed ~88% HN accuracy for this path while live inference showed ~20%: the two are not measuring the same thing.

Fix: once the CNN’s HN gate fires, trust it — the RF no longer gets a vote. This raised win-reason accuracy on the 6 matches tested before/after from 62.9% to 82.1%, and across the full 8-match set from 68.7% to 85.7%, with winner accuracy essentially unchanged (67.2% → 68.9%, since the override rarely flipped the *winner*, only the *reason*).

Known remaining issue: fault-side errors

69% of current winner errors have the correct `win_reason` but the wrong fault-side (i.e. wrong winner despite correctly classifying how the rally ended). This is unrelated to Strategy E and not yet investigated — the next place to look is the fault-side logic in `predict.py` (WBL uses shuttle-Y vs net floor line; HN/00B use the rule engine's geometric partition — see [06_win_prediction.md](#)).